

Josef Bisits

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EDUCATION

- **Doctor of Philosophy**, The University of New South Wales Sydney 2022–2025
School of Mathematics and Statistics
 - Supervisors: Associate Professor Jan Zika, Professor Trevor McDougall and Dr Taimoor Sohail
 - Thesis: [Non-linear controls on ocean circulation and mixing in the high-latitude oceans](#)
 - Published version of PhD thesis [abstract](#)
- **Bachelor of Science (Advanced mathematics, honours)**, The University of New South Wales Sydney 2016–2021
School of Mathematics and Statistics
 - Major: Applied mathematics
 - Supervisors: Associate Professor Jan Zika and Dr Geoff Stanley
 - Thesis: Passive tracer mixing along isopycnal surfaces in turbulent flows
 - Honours grade: class 1
- **Bachelor of Music**, The University of Sydney Sydney 2007–2010
Sydney Conservatorium of Music
 - Major: Performance, double bass
 - Double bass instructor: Alex Henery

SCIENTIFIC EXPERIENCE

RESEARCH POSITIONS

- **Postdoctoral research associate**, Australian Centre for Excellence in Antarctic Science UNSW, Sydney 2025-current
Diagnosing numerical mixing in finite volume models of fluid flow with A/Prof. Jan D. Zika.
- **Research assistant**, School of Mathematics and Statistics UNSW, Sydney 2022
Numerical modeling and analysis of tracer release experiments in turbulent flows with A/Prof. Jan Zika.
- **Research assistant**, Faculty of Built Environment UNSW, Sydney 2022
Historical water quality data analysis to develop a predictive water quality model with A/Lec. Simon Lloyd.

INDUSTRY POSITIONS

- **Ocean modelling contractor** [Submarine](#)
Build and validate a model to simulate tracer release experiments in Port Angeles Harbour, Washington, USA. 2025-2026

PUBLICATIONS

- **Bisits, Josef I.** et al. “A simple formula to infer rates of numerical mixing in finite volume ocean models part two: application”. In prep.
- Zika, Jan D., **Bisits, Josef I.** et al. “A simple formula to infer rates of numerical mixing in finite volume ocean models part one: theory and example”. In prep.
- **Bisits, Josef I.**, McDougall, Trevor J., and Zika, Jan D. Aug. 2026. “Determining how nonlinearities in the equation of state affect diffusive interfaces in the ocean”. In: *Journal of Fluid Mechanics* 1040, A33. DOI: [10.1017/jfm.2026.11889](https://doi.org/10.1017/jfm.2026.11889).

- Lloyd, Simon D., **Bisits, Josef I.** et al. 2026c. “Swimming in urban estuaries: understanding stormwater contamination events and recovery from historical data”. In: *Water Research*, p. 125686. DOI: [10.1016/j.watres.2026.125686](https://doi.org/10.1016/j.watres.2026.125686).
- **Bisits, Josef I.**, Zika, Jan D., and Sohail, Taimoor. May 2025. “Cabbeling as a catalyst and driver of turbulent mixing”. In: *Journal of Fluid Mechanics* 1011, A17. DOI: [10.1017/jfm.2025.349](https://doi.org/10.1017/jfm.2025.349).
- **Bisits, Josef I.**, Zika, Jan D., and Evans, Dafydd Gwyn. Dec. 2024. “Does Cabbeling Shape the Thermohaline Structure of High-Latitude Oceans?” In: *Journal of Physical Oceanography* 54 (12), pp. 2419–2430. DOI: [10.1175/JPO-D-24-0061.1](https://doi.org/10.1175/JPO-D-24-0061.1).
- **Bisits, Josef I.**, Stanley, Geoffrey J., and Zika, Jan D. Feb. 2023. “Can We Accurately Quantify a Lateral Diffusivity from a Single Tracer Release?” In: *Journal of Physical Oceanography* 53 (2), pp. 647–659. DOI: [10.1175/JPO-D-22-0145.1](https://doi.org/10.1175/JPO-D-22-0145.1).

SELECTED CONFERENCE PROCEEDINGS AND SEMINARS

ORAL PRESENTATIONS

- **Bisits, Josef I.**, Zika, Jan D., and Sohail, Taimoor. Feb. 2026. “Cabbeling as a catalyst and driver of turbulent mixing”. In: *Ocean Sciences*.
- **Bisits, Josef I.** Oct. 2025. “Non-linear controls on ocean circulation and mixing in the high-latitude oceans”. In: *Australis Centre for Excellence in Antarctic Science seminar series*.
- **Bisits, Josef I.**, Zika, Jan D., and Evans, Dafydd Gwynn. June 2024. “Does cabbeling shape the thermohaline structure of high-latitude oceans?” In: *Gordon Research Seminar on Ocean Mixing*.
- **Bisits, Josef I.**, Zika, Jan D., and Evans, Dafydd Gwynn. Feb. 2024. “Does cabbeling shape the thermohaline structure of high-latitude oceans?” In: *Australian Meteorological and Oceanographic Society annual conference*.
- **Bisits, Josef I.**, Stanley, Geoffrey J., and Zika, Jan D. July 2023. “Can we accurately quantify a lateral diffusivity from a single tracer release?” In: *XVIII General Assembly of the International Union of Geodesy and Geophysics (IUGG)*.
- **Bisits, Josef I.**, Stanley, Geoffrey J., and Zika, Jan D. Nov. 2022. “Can we accurately quantify a lateral diffusivity from a single tracer release?” In: *Australian Meteorological and Oceanographic Society annual conference*.

POSTER PRESENTATIONS

- **Bisits, Josef I.**, Zika, Jan D., and Sohail. June 2024. “The effect of non-linear processes on mixing”. In: *Gordon Research Conference on Ocean Mixing*.
- **Bisits, Josef I.**, Zika, Jan D., and Sohail, Taimoor. Nov. 2024. “The effects of cabbeling on mixing and energetics in polar oceans”. In: *Australian Antarctic Research Conference*.
- **Bisits, Josef I.**, Zika, Jan D., and Evans, Dafydd Gwynn. July 2023. “Does cabbeling shape the thermohaline structure of high-latitude oceans?” In: *Physics of the Ocean Summer School*.
- **Bisits, Josef I.**, Stanley, Geoffrey J., and Zika, Jan D. Nov. 2022. “Can we accurately quantify a lateral diffusivity from a single tracer release?” In: *Australian Centre for Excellence in Antarctic Science workshop*.

PROGRAMMING SKILLS

I have a wide experience in programming which includes:

- using High Performance Computing systems to run models on both CPU’s and GPU’s;
- building models for ocean and fluid dynamics simulations; and
- data visualisation.

My programming language of choice is  **julia**. Some packages I am involved with are:

- Author [StaircaseShenanigans.jl](#), [TwoLayerDirectNumericalShenanigans.jl](#), [RasterHistograms.jl](#);

- Developer `PassiveTracerFlows.jl`; and
- Contributor `FourierFlows.jl`, `GeophysicalFlows.jl`, `Oceananigans.jl` amongst others.

I am also fluent with , , , , $\text{\LaTeX}$ and `git`.

TEACHING EXPERIENCE

MATHEMATICS

- **Tutoring** UNSW, Sydney
2021–Current
I have been involved in mathematics tutoring and marking since my honours year at UNSW including:
 - Mathematics Drop in Centre;
 - Mathematics 1A (MATH1131/1141);
 - Mathematics 1B (MATH1231/1241);
 - Numerical Methods and Statistics (MATH2089);
 - Introduction to Atmosphere and Ocean Dynamics (MATH2241);
 - Fluids, Oceans, and Climate (MATH3261-5285); and
 - Third year project (MATH3001) marker.
- **Student supervision** July-September 2026.
 - Krishiv Kukreja, research intern (co supervisor with A/Prof. Jan Zika)

MUSIC

- **Private and ensemble double bass tuition** Sydney
2013–2022
During my double bass career I tutored private students and youth orchestra ensembles including:
 - Sydney Youth Orchestra ensembles;
 - Penrith Youth Orchestra; and
 - Playerlink, run by the Sydney Symphony Orchestra.

SCHOLARSHIPS AND AWARDS

- Best applied mathematics talk at UNSW School of Mathematics and Statistics postgraduate conference. 2024
- Michael Tallis Research travel award. 2023
- Australian Government Research Training Program Scholarship. 2022–2025
- School of Mathematics and Statistics research scholarship top-up. 2022–2025
- Australian Centre for Excellence in Antarctic Research PhD student. 2022–2025
- Michael Bannigan scholarship for academic achievement. 2010
- Corina d'Hage string scholarship. 2008–2010
- Winner of Sydney Youth Orchestra concerto competition. 2006

SELECTED OUTREACH ACTIVITES

- **Helper for Girls do the Maths!**, *University of New South Wales*. 2022, 2023
- **Helper for Experience UNSW Science day**, *University of New South Wales*. 2022
- **Climate @ maths display for UNSW open day**, *University of New South Wales*. 2022, 2023, 2024
- **Helper for uDASH work experience workshop**, *University of New South Wales*. 2022

PROFESSIONAL ACTIVITIES

- **Peer reviewer** 2023-current
 - Carried out peer reviews of articles submitted to:
 - Journal of Geophysical Research: Oceans;
 - EGUSphere;
 - Ocean Modelling.
- **Conference session chair** UNSW, Sydney 2024
 - School of Mathematics and Statistics postgraduate conference.

ORCHESTRAL EXPERIENCE

PROFESSIONAL

- **Sydney Symphony Orchestra** Sydney 2009–current
 - Regular casual double bassist, full time contract musician in 2014, 2015 and September 2017–September 2018.
- **Australian Chamber Orchestra** Sydney 2013–current
 - Regular casual double bassist.
- **Opera Australia Orchestra** Sydney 2016–current
 - Regular casual double bassist.
- **Click Track Music** Sydney 2009–current
 - Regular casual double bassist.
- **Tasmanian Symphony Orchestra** Tasmania 2013–2018
 - Casual double bassist.
- **Auckland Philharmonia** Auckland 2013
 - Casual double bassist.
- **Ulster Orchestra** Belfast Winter 2012
 - Held a trial for double bass section leader.

TRAINING PROGRAMS

- **Emerging Artist**, Australian Chamber Orchestra Sydney 2013
 - Mentor: Maxime Bibeau.
- **Southbank Sinfonia** London 2011-2012
 - Principal double bassist, twice appeared as a soloist.
- **Sydney Symphony Fellowship**, Sydney Symphony Orchestra Sydney 2010
 - Also appeared with Sydney Sinfonia from 2008-2010.
- **Australian Youth Orchestra** Australia 2007–2008
 - Double bass section player.
- **Sydney Youth Orchestra** Sydney 2006–2008
 - Principal double bassist, appeared as double bass soloist in 2006.

REFERENCES

- **Associate Professor Jan Zika**, *University of New South Wales, Sydney*. Email: j.zika@unsw.edu.au
- **Dr Taimoor Sohail**, *Research Fellow, University of Melbourne*. Email: taimoor.sohail@unimelb.edu.au
- **Emeritus Professor Trevor McDougall**, *University of New South Wales, Sydney*. Email: trevor.mcdougall@unsw.edu.au
- **Dr Geoff Stanley**, *Canadian Centre for Climate Modelling and Analysis*. Email: geoff.stanley@ec.gc.ca